Surya Sasaank Yanamandra

ysuryasasaank@gmail.com | +91 63038 68148
github.com/Sasaank79 | linkedin.com/in/surya-sasaank-yanamandra/ | surya-sasaank-yanamandra.vercel.app/

SUMMARY:

Computer Science Graduate (2025) with experience in backend development and machine learning applications. Skilled in building RESTful APIs using Java and Python, and working with relational databases and caching systems. Strong foundation in SQL, problem-solving, and practical software development.

EDUCATION:

MVGR College of Engineering, Vizianagaram, India **2021 - 2025**
Bachelor of Technology in Computer Science and Engineering (GPA: 7.61/10)
Relevant Courses: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks.

Sri Chaitanya Junior College, India **2019 - 2021**
Intermediate Education (Marks: 949/1000)

SKILLS:

Programming & Frameworks: Java, Python, SQL, Flask, scikit-learn, REST APIs.
Databases & Cloud: PostgreSQL, Redis, AWS Lambda.
Core Concepts & Tools: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Git, Machine Learning.

PROFESSIONAL DEVELOPMENT & TRAINING:

Salesforce - Journey 2 Employment Program| Program Participant | On-site | Feb 2024 - Aug 2024

- Selected in the top 15% (49/360+) of applicants for training on Data Structures and Code Optimization; applied Agile methodologies to refine algorithms for time and space complexity in peer reviews.

PROJECTS:

Inventory Management System (Flask, PostgreSQL, JWT, SQL) **Oct 2025**

- Built a **RESTful inventory management API** using **Flask** and **PostgreSQL**, designing a **3NF normalized schema** across 4 relational tables and exposing 20+ API endpoints for products, suppliers, transactions, and analytics.
- Implemented **JWT-based authentication** and authorization with 35 automated tests achieving 96% code coverage, improving **API reliability** and **access control**.
- Developed **SQL analytics queries** using **JOIN, GROUP BY, HAVING, and indexing** to support low-stock alerts, inventory insights, and **transaction-based stock tracking** across 100+ seeded products.

URL Shortener (Java, PostgreSQL, Redis) **Dec 2025**

- Developed a **URL shortening service** using **Java, PostgreSQL, and Redis**, generating 6-character **Base62 short codes** capable of supporting 56+ billion unique URL combinations.
- Integrated **Redis cache-aside caching** with 24-hour TTL, enabling sub-millisecond redirect lookups for frequently accessed URLs while **reducing database read load**.
- Implemented **HTTP 302 redirection** and **click-tracking workflows**, using **atomic database operations** to maintain consistency and prevent cache race conditions.

Phishing Link Detection Using ML (Python, scikit-learn, XGBoost, Random Forest) **Nov 2024 - Apr 2025**

- Engineered **URL-based lexical and domain-level features** including URL length, special characters, and domain age, applying **SMOTE oversampling** and **feature scaling** to improve model generalization on **imbalanced datasets**.
- Trained and evaluated **stacking ensemble models** combining **Random Forest, XGBoost**, and **SVM-based classifiers**, achieving 94% accuracy and 91% precision in **phishing URL classification**.
- Integrated a **whitelist/blacklist validation mechanism** and developed reproducible **ML workflows** with documentation and deployment-ready preprocessing pipelines.

Serverless Student Success Prediction System (AWS Lambda, Docker, XGBoost, LightGBM, CatBoost) **July 2025**

- Developed a **serverless ML inference pipeline** using **AWS Lambda, Docker, and AWS ECR** to predict student academic outcomes on a dataset containing 4,424 records and 44 engineered features.
- Built a **4-model stacking ensemble** combining **XGBoost, LightGBM, CatBoost**, and **Logistic Regression**, achieving a weighted F1-score of **0.77** on multiclass predictions.
- Designed and deployed a containerized cloud workflow using **EC2, Docker, ECR, and Lambda container images**, enabling scalable **serverless inference** without dedicated infrastructure management.

CERTIFICATIONS:

PCAP - Programming Essentials in Python • NPTEL - Cloud Computing • Cisco CyberSecurity Essentials • Cisco Linux Essentials • Machine Learning Specialization (DeepLearning.AI)